Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/02/2025
PI Name:	Diaz Artiles, Ana Ph.D.		
Project Title:	Effects of Altered-Gravity on Perception and Bi-manual Coordination: Impacts on Functional Performance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	08/01/2020	**End Date:**	09/30/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:	Parabolic		
Flight Assignment:	NOTE: End date changed to 9/30/2026 per V. Lehman/NASA JSC (Ed., 10/8/25) NOTE: End date changed to 7/31/2025 per NSSC information (Ed., 9/22/24) NOTE: End date changed to 7/31/2024 per NSSC information (Ed., 6/15/22) NOTE: End date changed to 7/31/2022 per NSSC information (Ed., 7/6/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dunbar, Bonnie Ph.D. (Texas A&M University) Kennedy, Deanna Ph.D. (Texas A&M University)		
Grant/Contract No.:	80NSSC20K1499		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Many of the activities associated with spaceflight require individuals to use both limbs simultaneously to accomplish the task. Motor control, as well as visual performance and spatial orientation are disrupted by gravitational transitions between 1 G and 0 G, but very little is known about the sensorimotor deficits between 0 G and 1 G. The objective of this analog-based research effort is to investigate the impact of partial G-levels on bimanual coordination tasks that are operationally relevant for spaceflight. The same set of human subjects will participate in two different bimanual coordination tasks during parabolic flight, which will deliver G-levels of 0, 0.25, 0.5, 0.75, 1, and 1.8 G. Sensorimotor dose-response curves will be generated between bimanual coordination operational variables as a function of G-level, and G-thresholds (which indicate when performance decrements occur) will be determined. We will also quantify the risk associated with the use of a common motion sickness drug (promethazine) during bimanual coordination tasks. Results will provide critical information for current and future countermeasure development and in-flight prescriptions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project investigates the influence of gravity on bimanual coordination using a variety of altered-gravity analogs. Results will provide critical information for current and future sensorimotor-related countermeasures and in-flight prescription. In addition, this research effort has direct application to bimanual coordination tasks on Earth, for example during complex tasks that require a coordinated two-limb movement, as well as for rehabilitation purposes.
Task Progress:	We successfully conducted our parabolic flight campaign in June 2023, and we have been focusing on data analysis and publications since then. During this past year, we have submitted two manuscripts for peer-review, and we have several more in different stages of preparation. The manuscripts already submitted for review are focused on the effects of promethazine on motion sickness in altered gravity and the neurophysiological effects of partial gravity on bimanual control. In addition, we keep analyzing our measures of coordination, motion sickness, and EMG-EMG coherence to identify significant factors altering performance and generating dose-response curves. We also continue to investigate the bimanual coordination disruption (and adaptation) following gravitational transitions experienced during the parabolic flight, and a manuscript is in preparation. To date, the core findings reveal that altered gravity significantly disrupts motor control. Results showed that at lower gravity levels, force output was reduced, temporal dynamics became less harmonic, and variability increased, particularly in complex coordination tasks (e.g., 90° and 1:2 multifrequency patterns). Electromyographic analysis using EMG-EMG cross-wavelet coherence identified a significant reduction in beta-band (13–30 Hz) neural coupling in 0g compared to 1g, suggesting weakened neural synchronization in the absence of gravitational input. Partial gravity levels partially restored both neuromuscular coupling and force coordination, highlighting the role of proprioceptive feedback in supporting motor control. Motion sickness was assessed concurrently using subjective ratings collected pre-, in-, and post-flight. Participants receiving placebo reported greater symptom severity, particularly in 0g, reinforcing the provocative nature of microgravity on vestibular and sensory systems. Promethazine was effective in attenuating these symptoms across all gravity levels, although mild side effects such as drowsiness were noted. Crucially, the influence of promethazine on motor performance revealed nuanced trade-offs. While the drug generally reduced force variability and prevented excessive force production—potentially stabilizing motor output in simpler tasks (e.g., 0° and 180° coordination)—it also impaired the system's capacity to adapt to gravitational transitions. In complex coordination tasks, promethazine was associated with elevated inter-peak interval variability and reduced timing precision, suggesting that the medication may interfere with cognitive-motor integration and adaptive sensorimotor strategies. Taken together, this investigation demonstrates that, while partial gravity can partially restore disrupted motor function in microgravity, pharmacological mitigation of motion sickness introduces its own performance limitations. These findings have important implications for astronaut training, countermeasure design, and operational planning, particularly for missions requiring fine motor skills under varying gravity conditions. Optimizing performance during spaceflight may require an integrated approach that balances motion sickness management with the preservation of neuromuscular adaptability and task-specific coordination capabilities. During this past year, we have also continued our data analysis for our ground experiment efforts. We have published an additional journal paper on the effects of display type and body position on motor performance (Weinrich et al. 2025), and additional manuscripts are in preparation. -Weinrich M, Neto OP, Wang Y, Abbott R, Walsh M, Diaz-Artiles A, Kennedy D. “Accuracy and stability in non-immersive VR: How display type and body position influence motor performance.” Hum Mov Sci. 2025 Jun;101:103360. During next year, we will continue our investigation using centrifugation. In addition, as part of a student augmentation grant received in January 2025, we will continue additional groundwork to investigate the use of lower body negative pressure (LBNP) as a tool to simulate gravitational transitions and associated disruption in bimanual coordination performance. The IRB for this upcoming study has been submitted and we are awaiting approval to start data collection.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Kennedy DM, Weinrich M, Abbott R, Neto O, Keller N, Wright T, Dunbar BJ, Diaz-Artiles A. "Synchronization in space: Exploring bimanual skills on the vomit comet." 2024 North American Society for the Psychology of Sport and Physical Activity (NASPSPA) Conference, New Orleans, Louisiana, June 5-8, 2024. Abstracts. 2024 North American Society for the Psychology of Sport and Physical Activity (NASPSPA) Conference, New Orleans, Louisiana, June 5-8, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Weinrich M, Wang Y, Abbott R, Diaz-Artiles A, Kennedy D.M. "Virtual reality displays mitigate head orientation effects on coordination dynamics: Insights from a bimanual force coordination study." 2024 North American Society for the Psychology of Sport and Physical Activity (NASPSPA) Conference, New Orleans, Louisiana, June 5-8, 2024. Abstracts. 2024 North American Society for the Psychology of Sport and Physical Activity (NASPSPA) Conference, New Orleans, Louisiana, June 5-8, 2024. , Jun-2024

Abstracts for Journals and Proceedings	Kennedy D, Neto OP, Weinrich M, Abbott R, Rao R, Patanam S, Keller N, Wright T, Dunbar BJ, Diaz-Artiles A. "Neurophysiological adaptions to altered-gravity: Bimanual coordination during parabolic flight." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Articles in Peer-reviewed Journals	Weinrich M, Neto OP, Wang Y, Abbott R, Walsh M, Diaz-Artiles A, Kennedy D. "Accuracy and stability in non-immersive VR: How display type and body position influence motor performance." Hum Mov Sci. 2025 Jun;101:103360. https://doi.org/10.1016/j.humov.2025.103360 ; PubMed PMID: 40349492 , Jun-2025
Dissertations and Theses	Keller N. "Linking cardiovascular and neuromotor responses to an orthostatic challenge." Ph.D. Dissertation, Kinesiology, Texas A&M University, May 2024. , May-2024
Dissertations and Theses	Wang, Y. "Hemispheric influence of learning and consolidation of a dynamic patters with 90 degree relative phase." Ph.D. Dissertation, Kinesiology, Texas A&M University, August 2023. , Aug-2023